CASE STUDY

Adversarial Robustness and Cat-to-Object Perception Gap

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Abstract

Deep convolutional neural networks achieve remarkable accuracy on standard benchmarks, yet remain brittle in a specific, measurable way. This study empirically characterizes that brittleness by identifying the minimum L∞-bounded perturbation magnitude (ε) required to force a targeted misclassification from a biological entity (Cat, classes 281–285) to an inanimate textile (Rug, class 762). A pre-trained ResNet-50 model can be forced to misclassify every image of a cat as a rug with 99%+ confidence using pixel perturbations so small they are completely invisible to the human eye. This study maps the precise boundary, $\varepsilon \approx 0.015$, where human perception and machine vision permanently diverge. Starting with the standard Fast Gradient Sign Method (FGSM), I observed that single-step attacks scatter predictions unpredictably across adjacent classes rather than converging on a chosen target. Upgrading to Iterative FGSM (I-FGSM) solved this enabling precise, surgically directed misclassification. The result is a complete empirical map of the Perception Gap i.e. the narrow ε band in which a human sees a normal photograph while the model registers something else entirely, and does so with near absolute confidence.

1 Introduction

Neural networks operating in high dimensional pixel spaces have a well-documented vulnerability which is, they are sensitive to mathematically derived noise patterns that are perceptually meaningless to humans. *Szegedy et al. (2013)* [2] first demonstrated that imperceptible input perturbations could cause large changes in network outputs, and *Goodfellow et al. (2014)* [1] provided the theoretical underpinning via the linearity hypothesis i.e. the observation that small perturbations accumulate into large inner product shifts across many dimensions. This particular study frames the vulnerability in concrete, interpretable terms. Confusing a living animal with a woven textile is semantically extreme. The kind of misclassification that underscores how divorced a model's internal representation can be from the visual semantics a human would use. In safety-

critical contexts like autonomous vehicles, medical imaging, surveillance, that semantic gap can have serious consequences.
The goal here is not merely to demonstrate that adversarial examples exist, but to measure them precisely to find exactly where the model breaks, how much perturbation it takes, and what the attack trajectory looks like as ε increases from zero.

2 Methodology

2.1 Architecture and Preprocessing

The target model is ResNet-50 with weights pre-trained on ImageNet-1K, loaded without modification via PyTorch. All input images were resized to 224 × 224 pixels, converted to tensors in the range [0, 1], and normalized using the standard ImageNet channel-wise mean (μ = [0.485, 0.456, 0.406]) and standard deviation (σ = [0.229, 0.224, 0.225]). This normalization step is critical as adversarial perturbations are generated in normalized space and must be scaled consistently to remain within the stated ε bound.
The test corpus was a curated set of 7 high-resolution images, each cleanly classified as a cat breed (classes 281-285) at $\varepsilon = 0$. Images with confounding backgrounds, for example, a cat partially obscured inside a plastic bag were excluded after initial testing showed that visual clutter introduced ambiguity into baseline predictions, making it impossible to isolate the effect of the attack itself.

2.2 Attack Design: From FGSM to I-FGSM

The experiment began with standard FGSM, which constructs an adversarial example in a single step by moving the input in the direction of the loss gradient :

$$x_adv = x + \varepsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

FGSM reliably destroyed confidence in the 'Cat' class, but it acted as a blunt instrument as predictions scattered across neighboring textile classes (hammocks, mosquito nets, bath towels) rather than converging on the intended target. A single large gradient step does not navigate the high-dimensional decision boundary with enough precision for a targeted attack. The solution was to switch to Iterative FGSM (I-FGSM), a multi-step variant of Projected Gradient Descent. Rather than one coarse step, I-FGSM takes multiple micro-steps of size α, recalculating the gradient of the cross-entropy loss with respect to the target class at each iteration and projecting the result back into the valid ε-ball :

$$x_\{N+1\} = \text{Clip}_\{x,\varepsilon\} \{ x_N - \alpha \cdot \text{sign}(\nabla_x J(\theta, x_N, y_target)) \}$$

Parameters: $\alpha = 0.005$, **10 iterations**, total perturbation clamped to ε. The subtraction (rather than addition) in the iterative step is deliberate as the loss is being minimized with respect to the target class, steering the image toward the target's decision region rather than away from the source class.

2.3 Evaluation Protocol

Attacks were evaluated across eight ε values under the L∞ norm : {0, 0.001, 0.005, 0.01, 0.02, 0.05, 0.075, 0.1}. For each image at each ε, I recorded the model's top-1 predicted class, the confidence in the original class (Cat), and the confidence in the target class (Rug). Attack success was defined as top-1 prediction equaling class 762 (Rug).

3 Results

3.1 Degradation of the Original Class

As ε increases, the model's confidence in 'Cat' does not decay gradually, it collapses. The degradation curve is non-linear and exhibits a clear threshold behavior. Across all seven images, the model retained near-baseline confidence up to ε = 0.005, then lost the majority of its original class confidence between ε = 0.005 and ε = 0.01. By ε = 0.01, no image in the corpus was still classified as a cat.
This threshold collapse is itself a meaningful finding. It demonstrates that the model's learned representation of 'cat' is not smoothly degradable, it is fragile at a specific perturbation magnitude and robust below it, which creates a sharp, exploitable boundary.

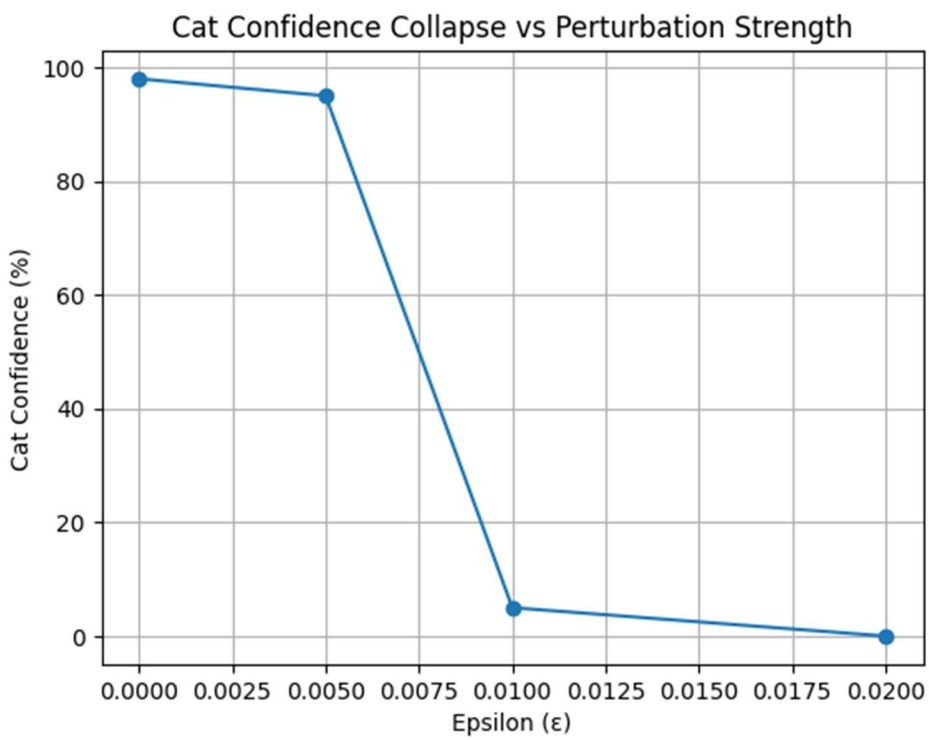

Figure 1 : Confidence Degradation - Model confidence in the true class (Cat) as ε increases from 0 to 0.1. Note the non-linear threshold collapse near ε = 0.01.

3.2 Targeted Ascent and the Texture Hypothesis

The other side of this transition is the rise of the adversarial target. As Cat confidence collapses, Rug confidence does not merely increase, it accelerates. In the clearest example from the corpus (cat-pic-7), target confidence increased from 17.8% at ε = 0.005 to 94.3% at ε = 0.01, and plateaued at 99.9% by ε = 0.02. That is an 82-percentage-point increase over a perturbation increment that remains visually imperceptible.

This behavior is consistent with the Texture Hypothesis (*Geirhos et al., 2018*),[3] which proposes that CNNs rely predominantly on local high-frequency texture features rather than global structural shapes. The I-FGSM attack, by repeatedly following the gradient of the target class loss, effectively superimposes an invisible 'rug texture' signature onto the pixel array. The model responds to this texture signal with high confidence, entirely ignoring the global shape of a sitting or reclining cat.

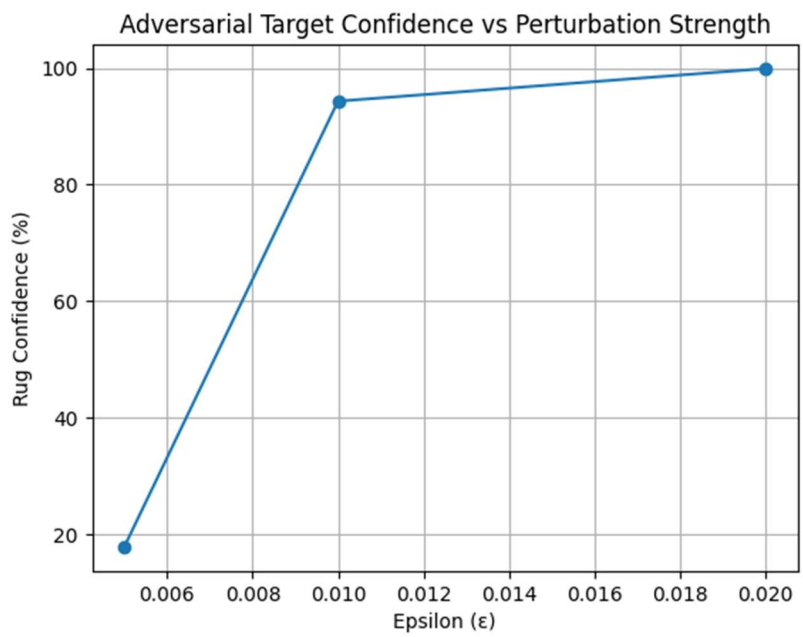

Figure 2 : Targeted Class Ascension — Model confidence in Rug (class 762) as ε increases. The exponential rise corresponds precisely to the collapse in Figure 1.

3.3 The Perception Gap — Empirical Definition

The central result of this study is the empirical mapping of the Perception Gap: the ε range in which the adversarial image remains visually indistinguishable from the original to a human observer, yet is classified by the model as a completely different semantic category with overwhelming confidence.

Phase	ε Range	Success Rate	Observation
Resistance	≤ 0.005	0%	Decision boundary holds
Breach	0.01	**71.4%**	Threshold collapse begins
Total Subversion	≥ 0.02	**100%**	All images misclassified at >99% confidence

Table 1 : Phase transitions in model behavior as perturbation magnitude (ε) increases. The Perception Gap is bounded between ε = 0.01 and ε = 0.02.

The gap defined here (ε) between 0.01 and 0.02, corresponding to roughly 1–2% pixel noise under L∞ is the operational danger zone. Within this range, the attack is already fully effective from the adversary's perspective, while remaining fully imperceptible from the defender's.

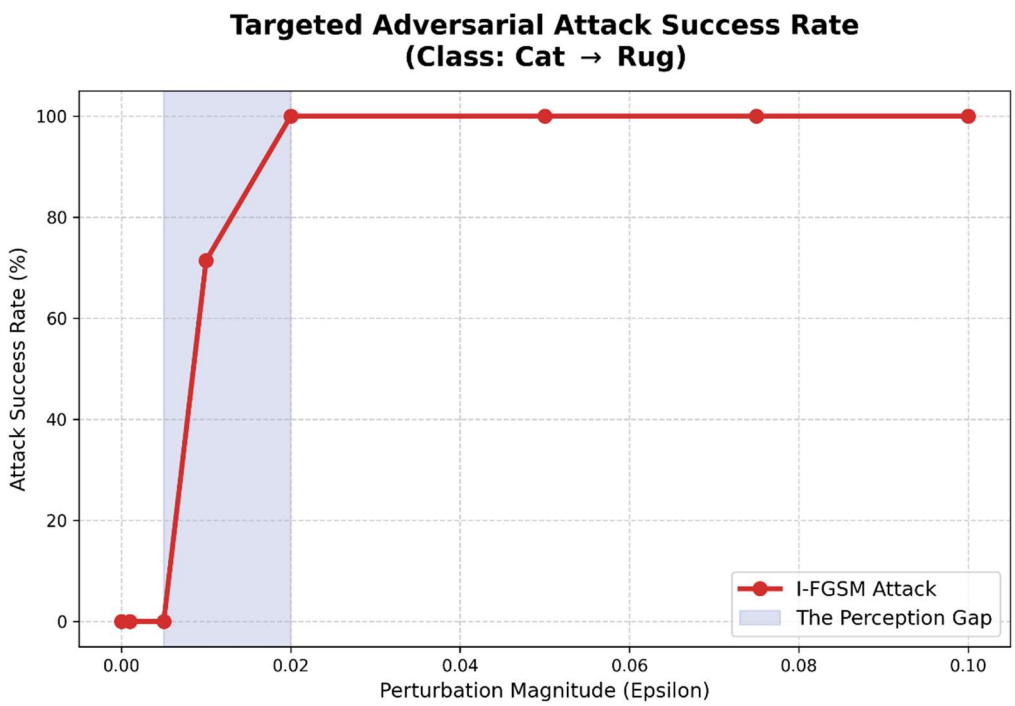

Figure 3: Attack Success Rate vs. ε — The Perception Gap visualized. The step function from 0% to 100% success rate occurs within the visually imperceptible perturbation range.

4 Discussion

What this experiment ultimately demonstrates is not just that adversarial examples exist, but that they are geometrically inevitable given how CNNs are trained. Standard cross-entropy loss on clean images teaches a model to find the fastest path to a correct classification, and that path runs through high-frequency texture features. The model never learns that a texture pattern divorced from global shape is meaningless because on the clean training distribution, it never needs to.
The transition from FGSM to I-FGSM was the most instructive step in the process. It illustrated a fundamental principle of navigating high-dimensional decision boundaries requires iterative refinement, not brute-force single steps. The shift from 'destroy the original class' to 'engineer a specific target class' is the difference between noise and a surgical instrument. Three classes of mitigation are worth noting [4], though none constitutes a complete solution. They are :

Adversarial Training - augmenting training data with adversarially perturbed examples, so the model learns to treat micro-gradient patterns as uninformative noise. It is effective but computationally expensive and often brittle against unseen attack types.
Input Sanitization - applying bit-depth reduction or spatial smoothing to inputs before inference, which degrades the high-frequency perturbation signal before it reaches the model. This one is simple but can reduce clean-image accuracy.
Shape-Biased Training - training on stylized-ImageNet or similar datasets that remove texture as a reliable cue, forcing the model to rely on structural shape. It is more fundamentally aligned with how human vision works, but requires significant dataset engineering.

Henceforth, none of these approaches fully closes the gap. Adversarial training in particular is prone to the cat-and-mouse dynamic where a stronger attack can usually be found once a defense is known. The more durable path is likely architectural as that explicitly encodes spatial invariance and shape-based reasoning, rather than relying on texture statistics.

5 Conclusion

Iterative FGSM achieves 100% targeted misclassification of cats as rugs at $\varepsilon \geq 0.02$, perturbations that are completely invisible to human observers. The Perception Gap ($\varepsilon \approx 0.015$) defines the precise threshold where human and machine visual reality permanently diverge. The Perception Gap is not an abstract concept. It is a measurable, reproducible, exploitable property of every CNN trained with standard methods on standard data. This study has characterized it empirically for one specific model, one specific class transition, and one specific attack family but the underlying mechanics generalize.
What I found most striking is how sharp the transition is. There is no gradual erosion. The model is confidently correct, then at a specific, small ε, it becomes confidently wrong. That sharpness suggests the decision boundary has a geometric structure that a one-step attack misses but an iterative attack finds precisely. Understanding that geometry is, in my view, the most important open question in adversarial robustness research.

References

[1] Goodfellow, I. J., Shlens, J., & Szegedy, C. (2014). Explaining and harnessing adversarial examples. arXiv:1412.6572.

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[4] Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2017). Towards deep learning models resistant to adversarial attacks. arXiv:1706.06083.